Unit: 6 Unit 8: Quadratics **Name:** ________________
Topic: Definition of Quadratics and Standard Form **Date:** __________

Advanced (Habanero)

Q1. What is the standard form of a quadratic function?
(A) $ax^2 + bx + c$
(B) $ax + bx^2 + c$
(C) $ax^2 - bx + c$
(D) $ax^3 + bx^2 + c$
(E) $ax^2 + b$

Q2. Which of the following is a quadratic function in standard form?
(A) $x^2 + 3x - 2$
(B) $x^3 - 2x^2 + x$
(C) $x^2 - 4$
(D) $2x^2 + 5x$
(E) $x + 2$

Q3. What is the value of a in the quadratic function $2x^2+3x-1$?
(A) 1
(B) 2
(C) 3
(D) -1
(E) -2

Q4. Which of the following quadratic functions has a minimum value?
(A) $x^2 + 2x + 1$
(B) $-x^2 + 2x - 1$
(C) $x^2 - 2x + 1$
(D) $-x^2 - 2x - 1$
(E) $x^2 + 1$

Q5. What is the axis of symmetry of the quadratic function $x^2 + 2x + 1$?
(A) $x = -1$
(B) $x = 1$
(C) $x = -2$
(D) $x = 2$
(E) $x = 0$

Q6. Which of the following is a key feature of the quadratic function $x^2 + 2x + 1$?
(A) Minimum value
(B) Maximum value
(C) Axis of symmetry
(D) Vertex
(E) All of the above

Q7. What is the vertex of the quadratic function $x^2 + 2x + 1$?
(A) $(-1, 0)$
(B) $(-1, -1)$
(C) $(-2, 1)$
(D) $(-2, -1)$
(E) $(-1, 1)$

Q8. Which of the following quadratic functions has a maximum value?
(A) $x^2 + 2x + 1$
(B) $-x^2 + 2x - 1$
(C) $x^2 - 2x + 1$
(D) $-x^2 - 2x - 1$
(E) $-x^2 + 1$

Open Ended Questions (FRQ)

Q9. Describe the key features of the quadratic function $x^2 - 4x + 3$. Provide the equation of the axis of symmetry and the coordinates of the vertex.

Q10. Write a quadratic function in standard form with a minimum value of -2 and an axis of symmetry at $x = 1$. Provide a brief explanation of how you determined the function.

Chef's Tips

- Emphasize the importance of identifying the standard form of a quadratic function.
- Provide examples of real-world applications of quadratic functions.
- Encourage students to use graphing calculators to visualize quadratic functions.